The Potential Outcome Framework, Part 1

Stat 256, Causality

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Readings

Chapter 1-3 from Hernan & Robins, Causal Inference: What If.

Outline

1 Potential Outcomes

The Neyman-Rubin Potential Outcome Model

Much of the applied causal inference literature speaks the language of the Neyman-Rubin causal model, aka the Potential Outcomes Model (POM).

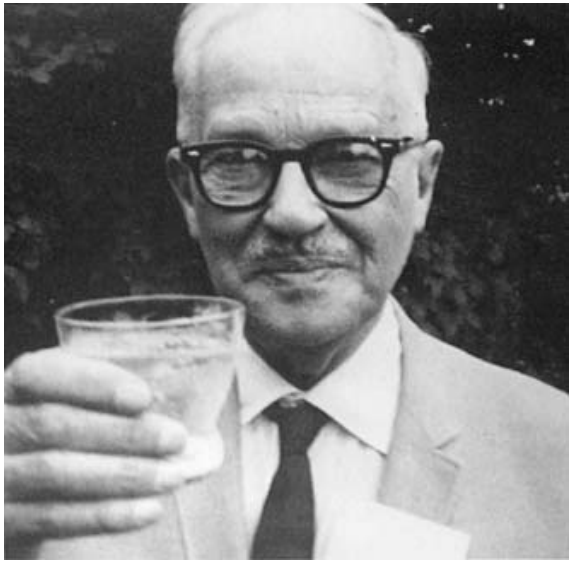

Figure: Neyman

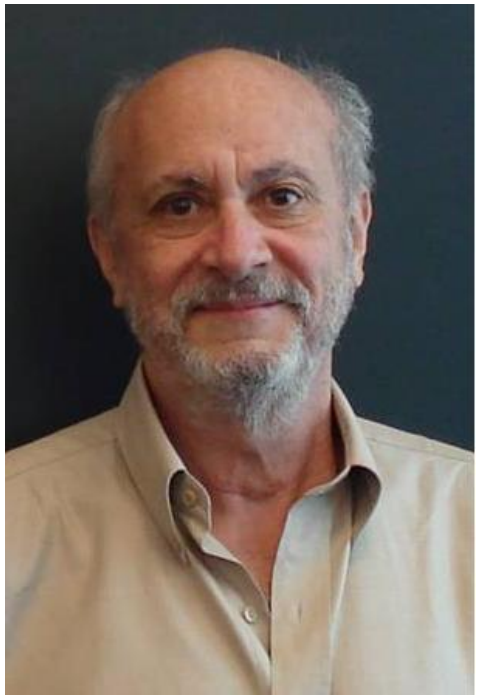

Figure: Rubin

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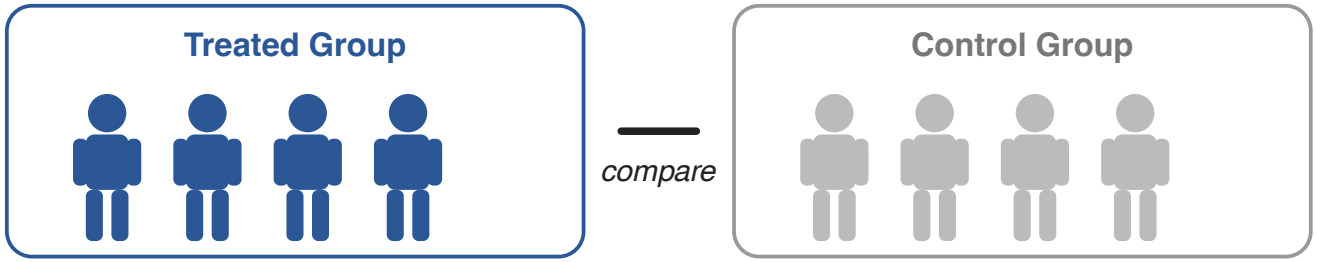
The first is a comparison *between groups of different units*. The second is a comparison *within each unit*, between two versions of its world.

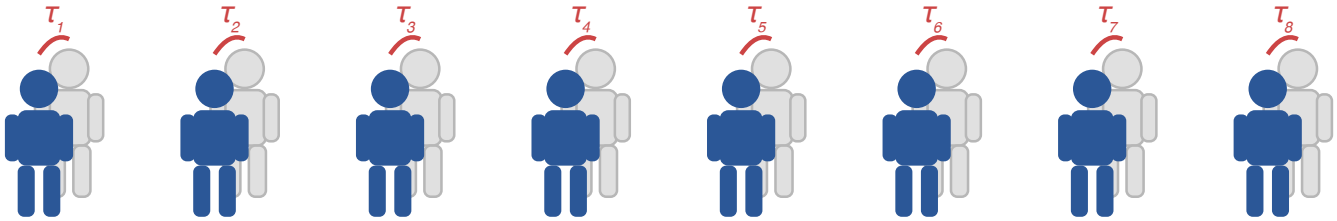
Group comparison vs. within-unit comparison

Group comparison



Contrasting two groups

Causal (counterfactual) comparison



Contrasting two (potential) outcomes for everybody

= treated = untreated τ_i = *individual contrast*

Potential Outcomes, Labeled

- Y_{1i}: the outcome unit i *would have* under treatment
- Y_{0i}: the outcome unit i *would have* under control
- $\tau_i = Y_{1i} - Y_{0i}$: the causal effect for unit i

Fundamental problem: we only ever see *one* potential outcome per unit.

From potential to observed

Adding the treatment indicator $D_i \in \{0, 1\}$, the observed outcome is

$$Y_i = D_i \, Y_{1i} + (1 - D_i) \, Y_{0i}.$$

Key idea: D_i doesn't <u>change</u> the potential outcomes — it just picks which one you see.

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Which of these are *always*, *sometimes*, or *never* observable?

- D_i $\quad Y_i$ $\quad Y_{1i}$ $\quad Y_{0i}$ $\quad \tau_i$

POs and QOIs

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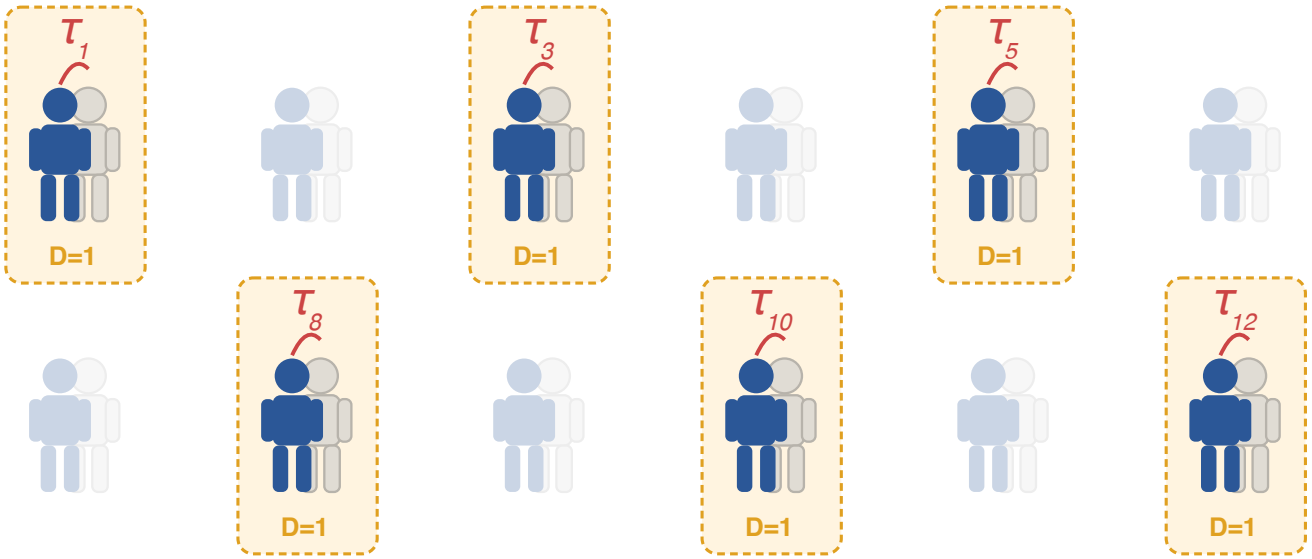
$$\mathbb{E}[\tau_i | D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0]$$

- Conditional average treatment effect (CATE):

$$\mathbb{E}[\tau_i | X = x] = \mathbb{E}[Y_{1i} - Y_{0i} | X = x]$$

Visualizing the ATT

The ATT asks: among units actually treated ($D_i = 1$), what's the average of their individual causal effects? Not a group comparison — imagine each treated unit's two potential lives:

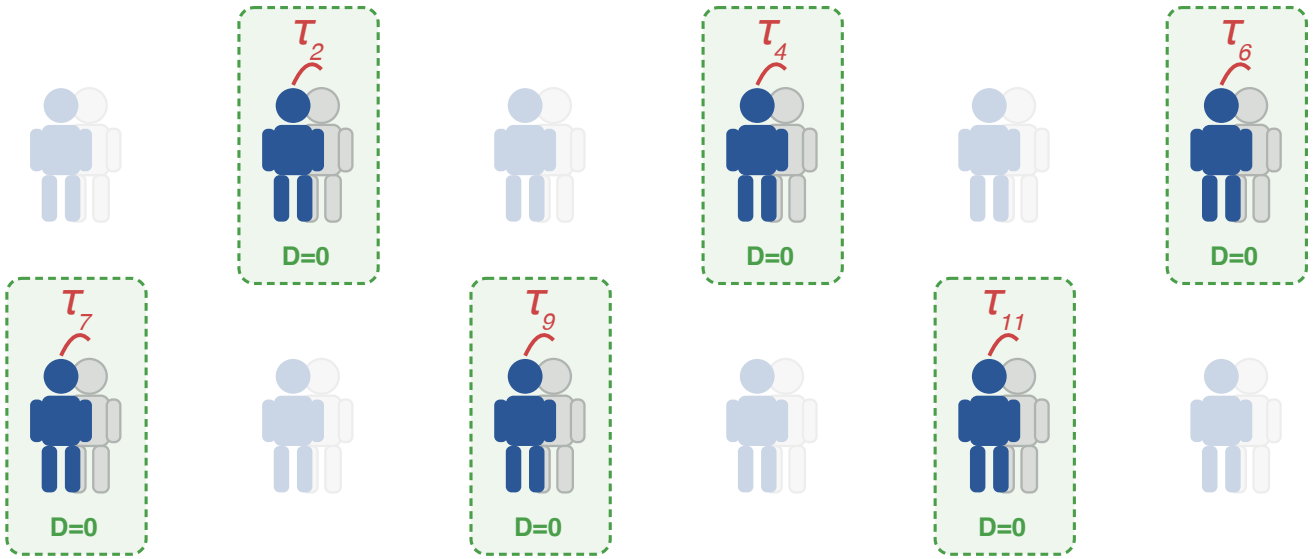


ATT = average of

$\tau_1, \tau_3, \tau_5, \tau_8, \tau_{10}, \tau_{12}$

Visualizing the ATC

The ATC asks the same question for the untreated ($D_i = 0$). These units were *not* treated, but we still ask what *would have happened* if they had been:



ATC = average of

$\tau_2, \tau_4, \tau_6, \tau_7, \tau_9, \tau_{11}$

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By writing $Y_i(d)$ we assert **consistency**: $Y_i(d)$ is the Y_i you'd see if i actually received treatment d. We name it so we can reference it later and notice where it breaks.

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("One version of treatment" is easy to overread; we'll save that for later.)

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A common guideline: a cause should be **manipulable** — you should be able to describe an ideal experiment that would set $D = d$.

- Helpful prodding, not a hard rule: non-manipulable things can have effects too, and the SCM can define those cleanly.

Brain Training

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- So can we estimate $\mathbb{E}[Y_{1i}]$? And $\mathbb{E}[Y_{0i}]$? How? What licenses do you use?

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- Can you say anything about expected direction of bias?

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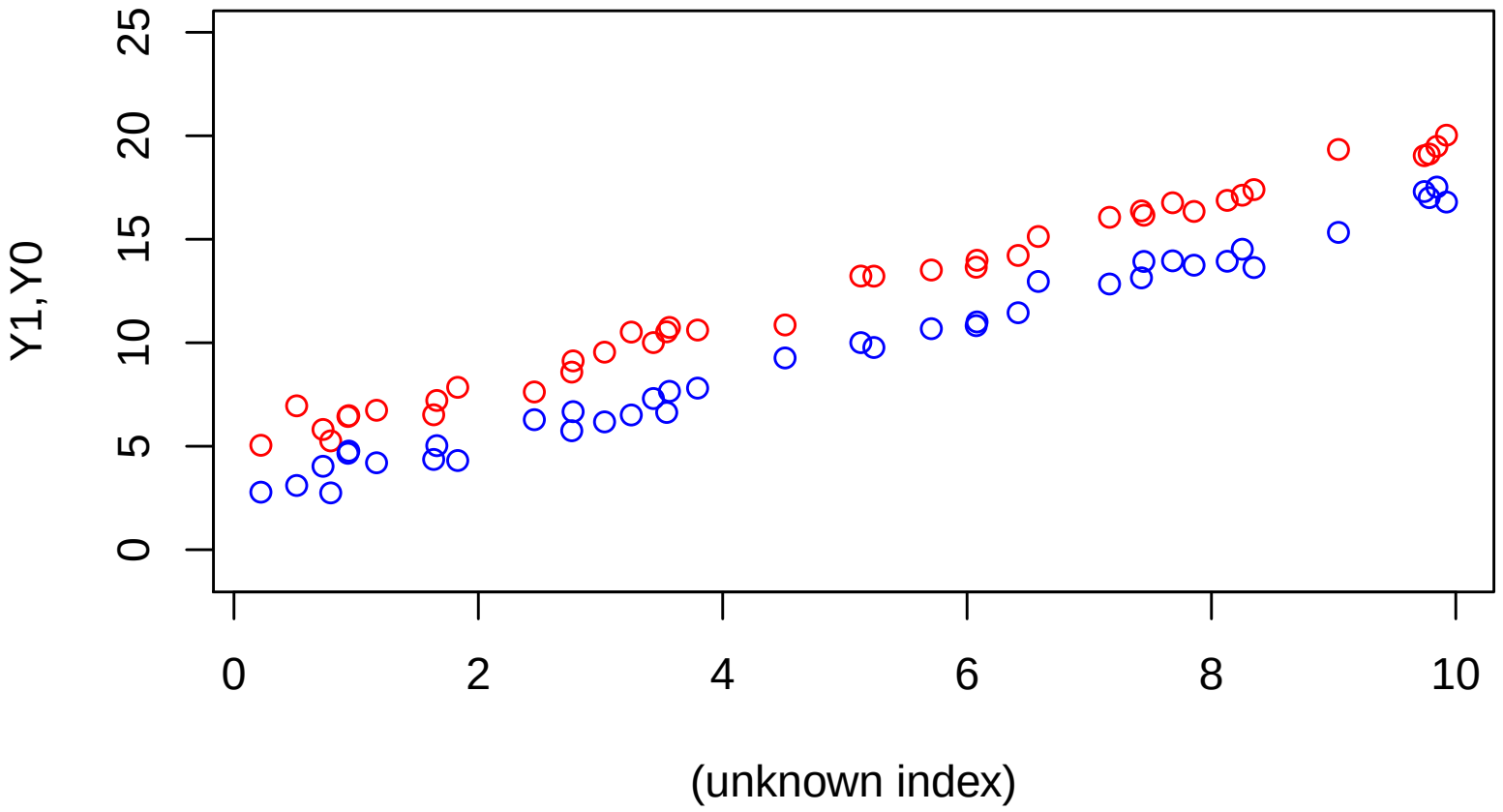
- Unfortunately, this is often the scenario we are in with observational data.

Brain Training: Visual Practice

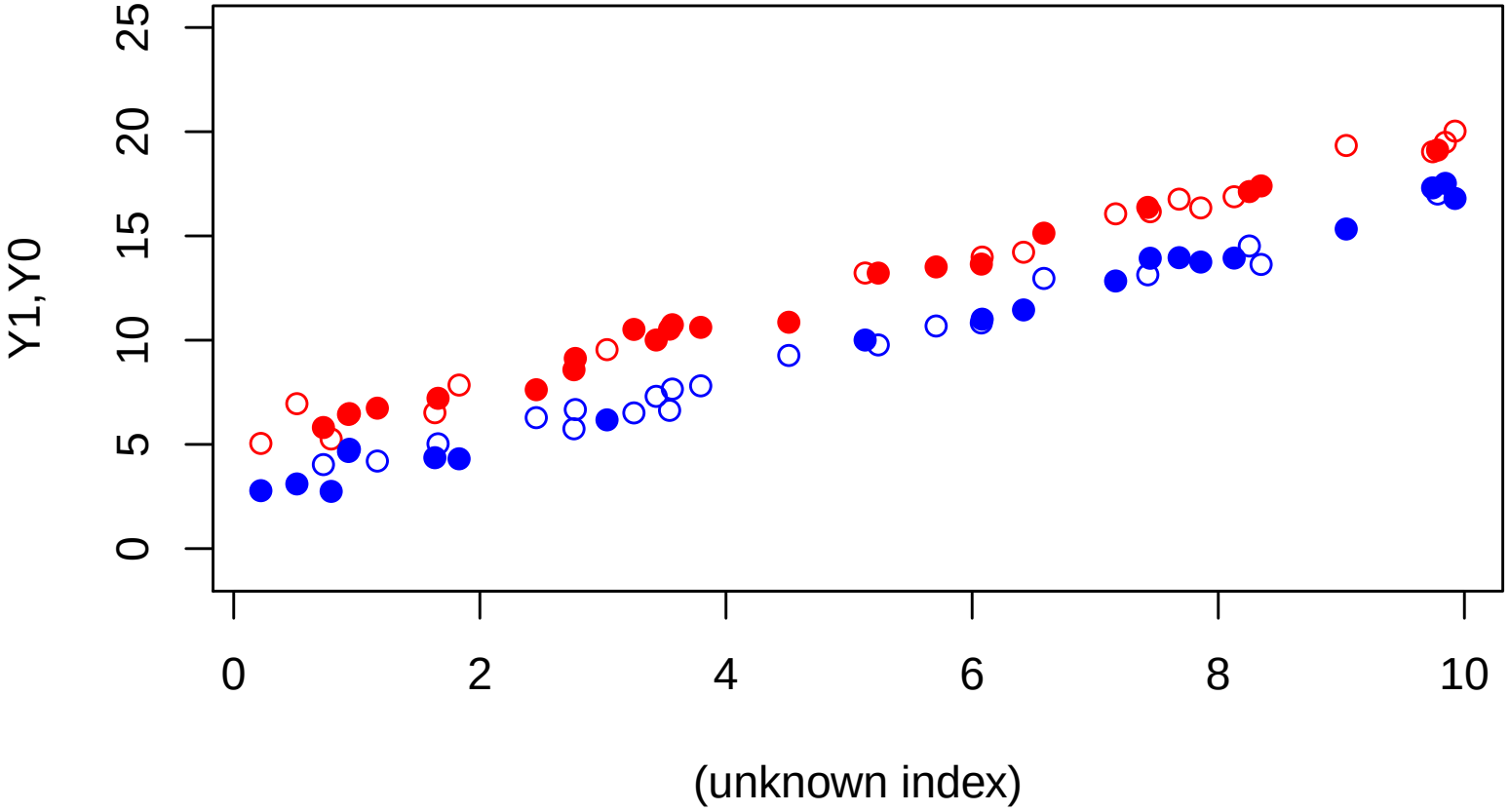
Let's look at Y_{1i} and Y_{0i} alone first.



True $\mathbb{E}[Y_{1i} - Y_{0i}] = \text{ATE} = 3$

Random Assignment of Treatment

Now suppose $D_i = 1$ is randomly assigned. We see only the filled dots:



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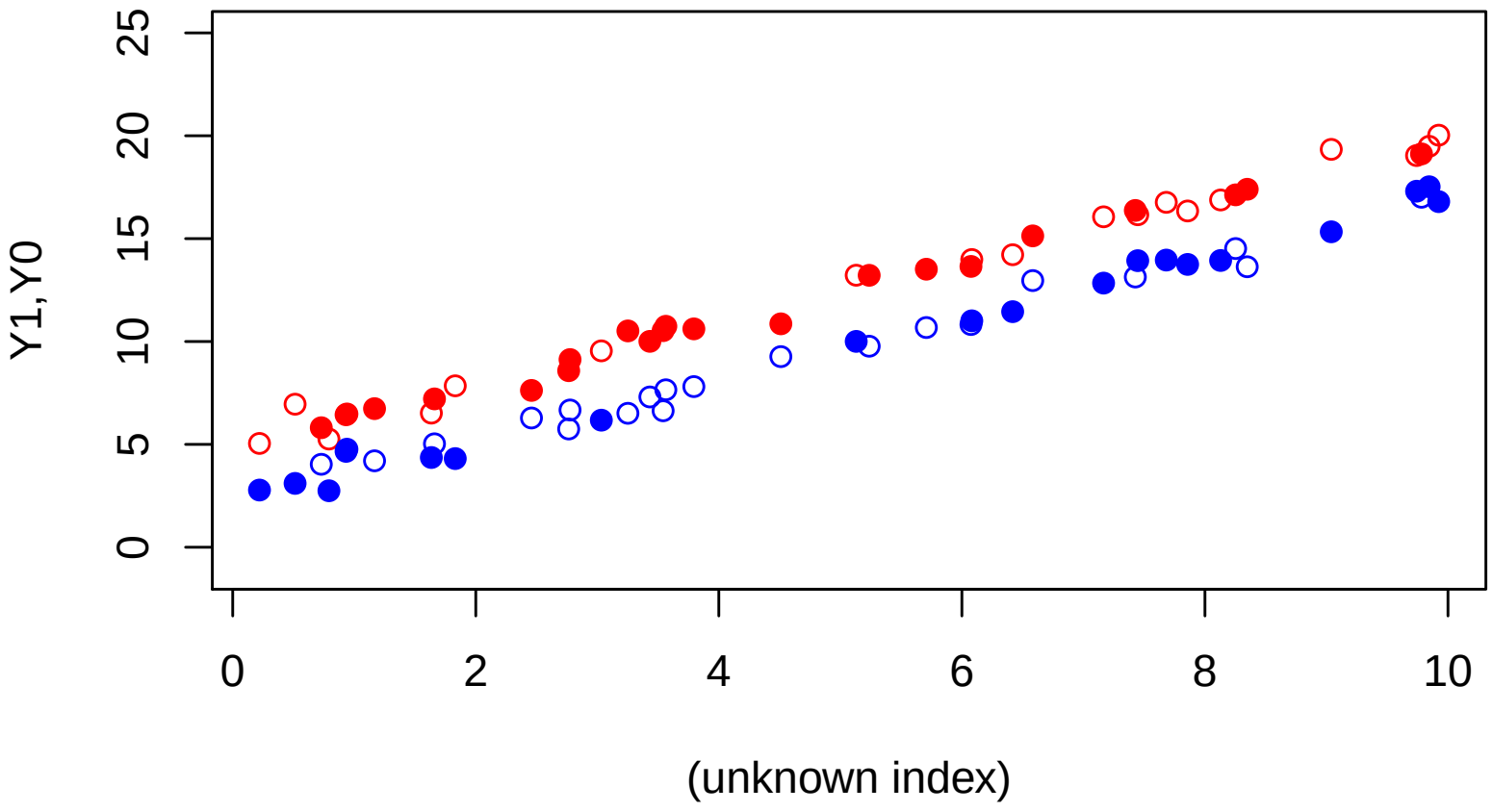
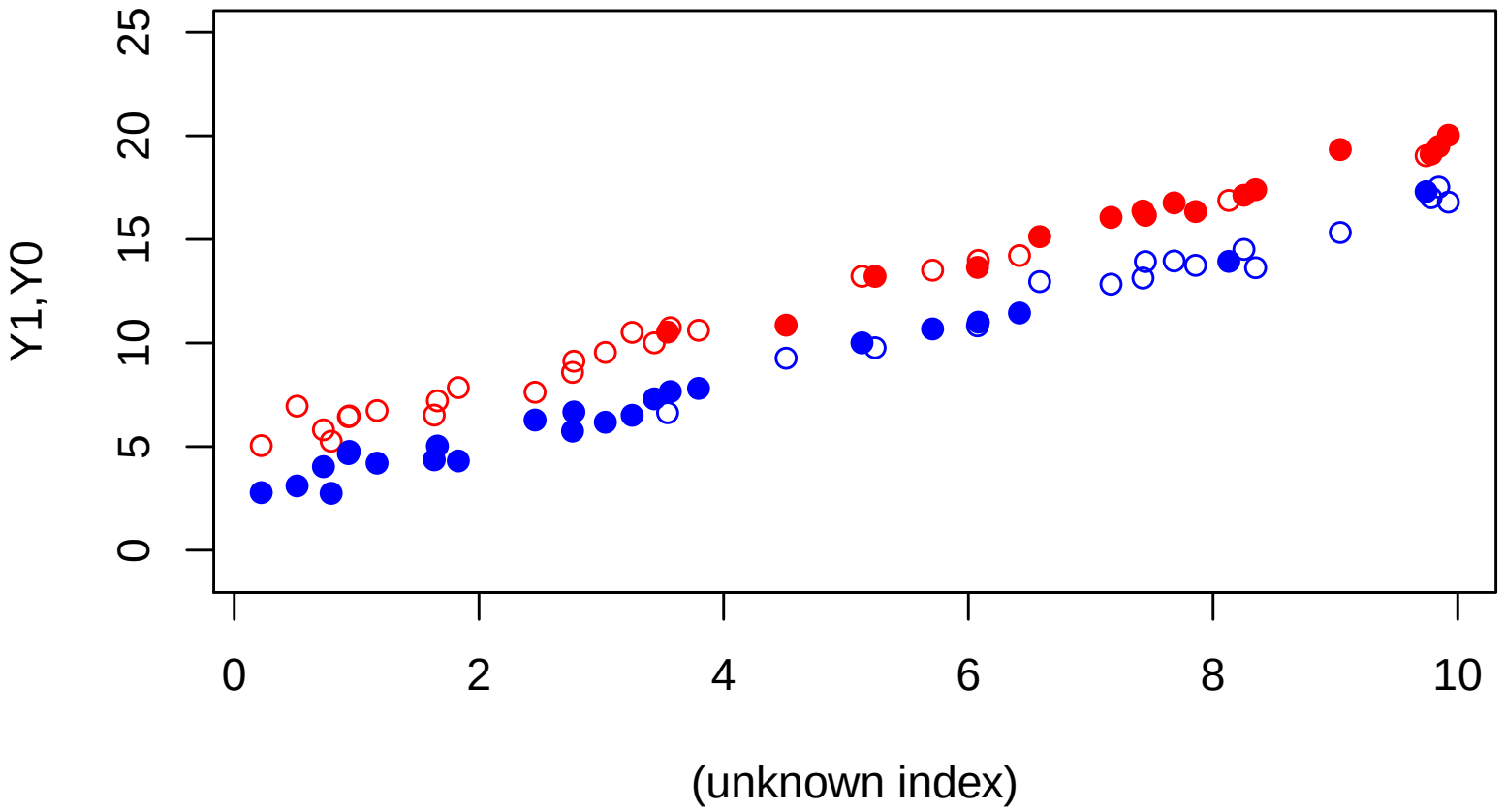


We get

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{\text{DIM}} = 3.01 \approx \text{ATE}$$

Selection into treatment, case 1

Suppose $Pr(D_i = 1)$ increases to the right. Now we observe (filled dots):



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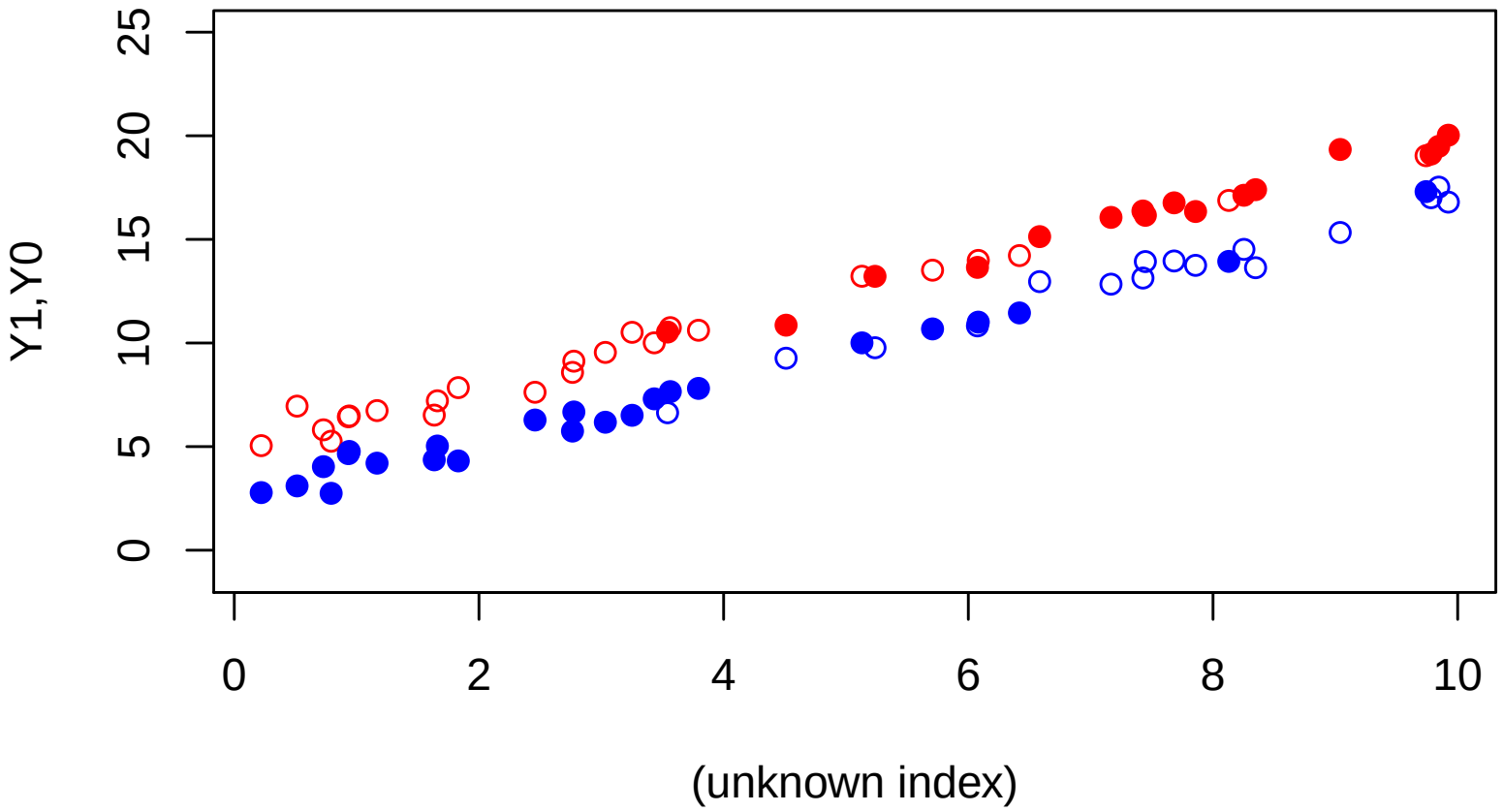

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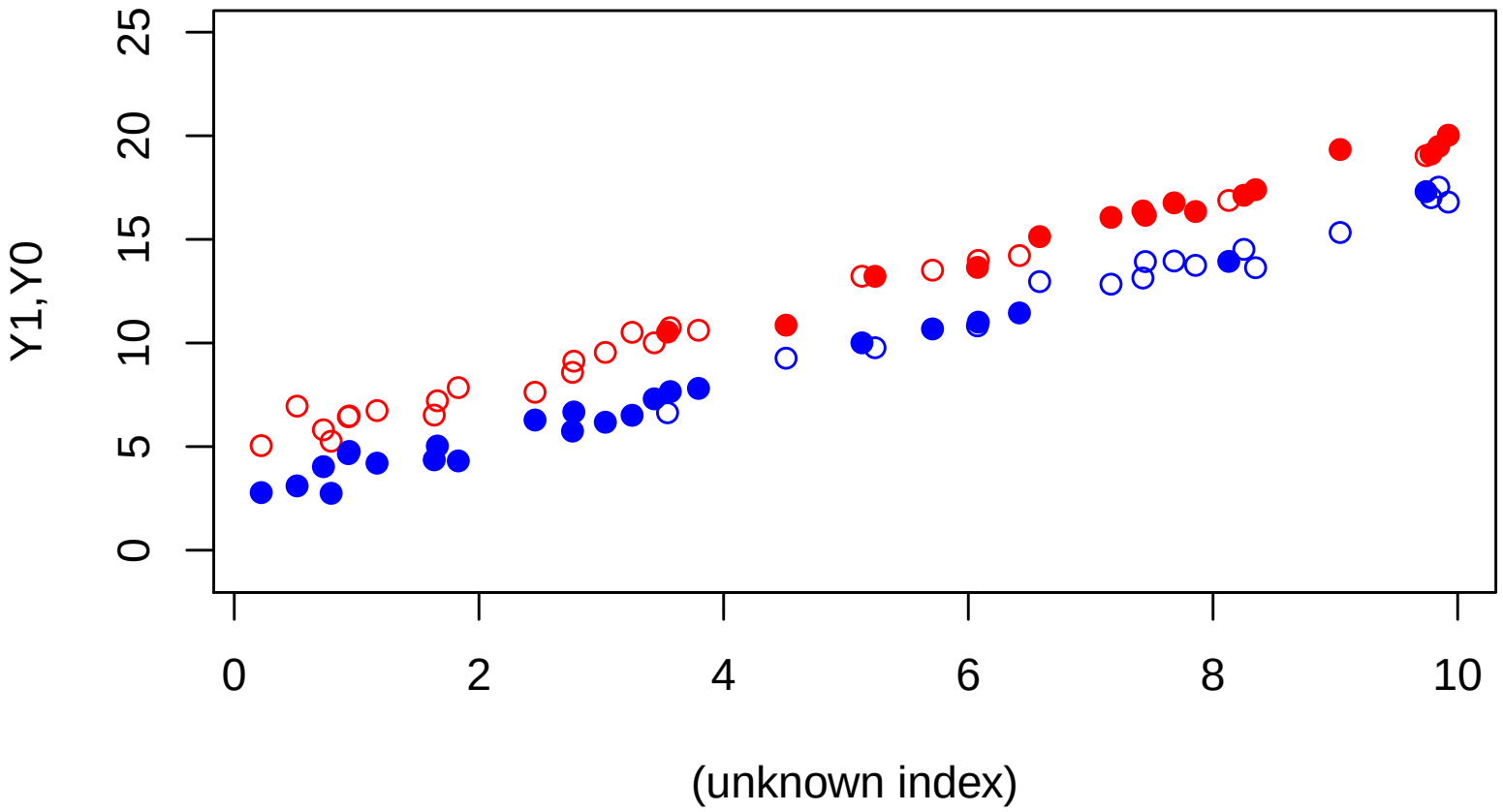

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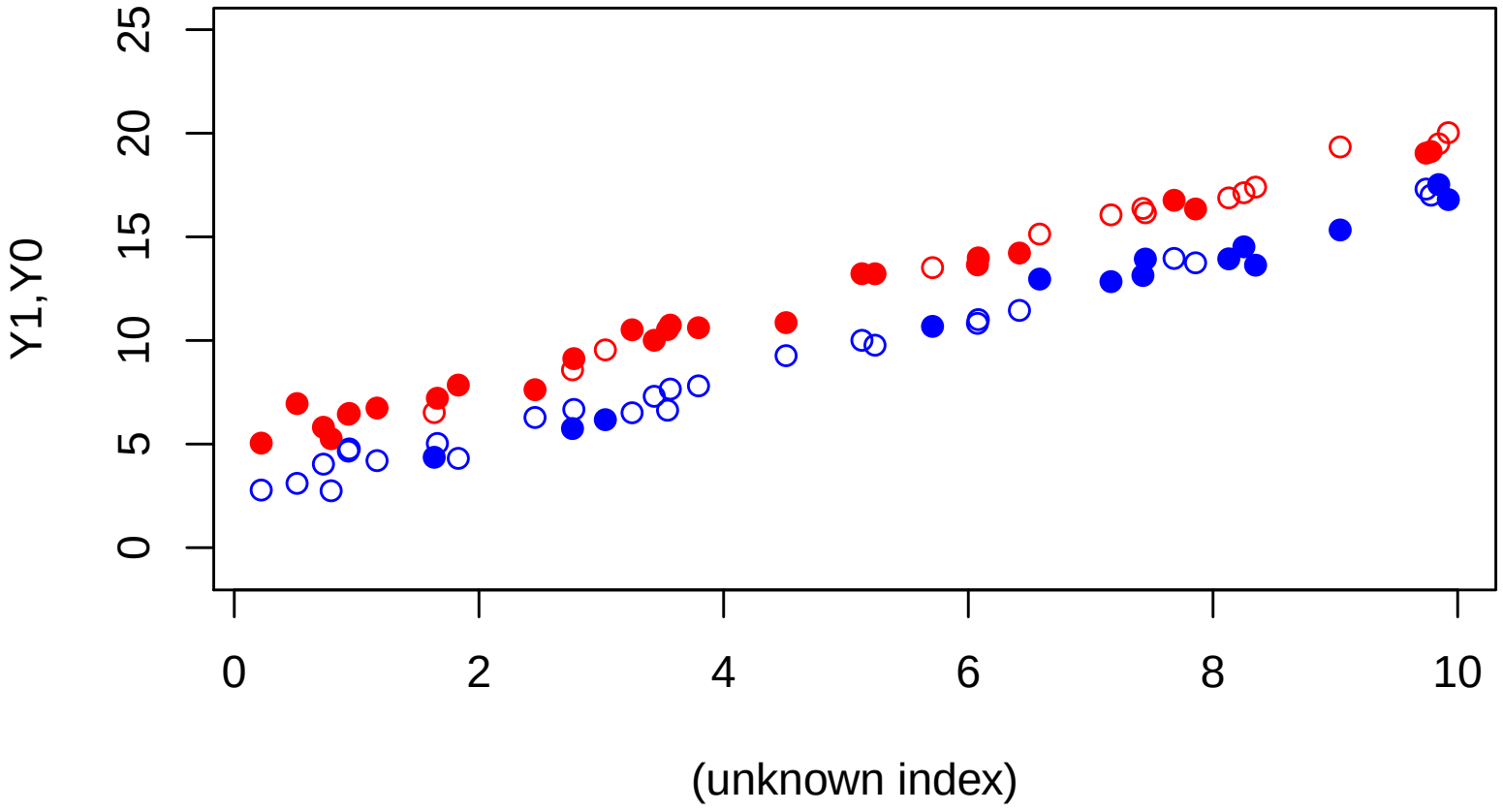
$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{\text{DIM}} = 9.97 \neq \text{ATE} = 3$$

Selection into treatment, case 2

Now suppose $Pr(D_i = 1)$ decreases to the right. We observe (filled dots):



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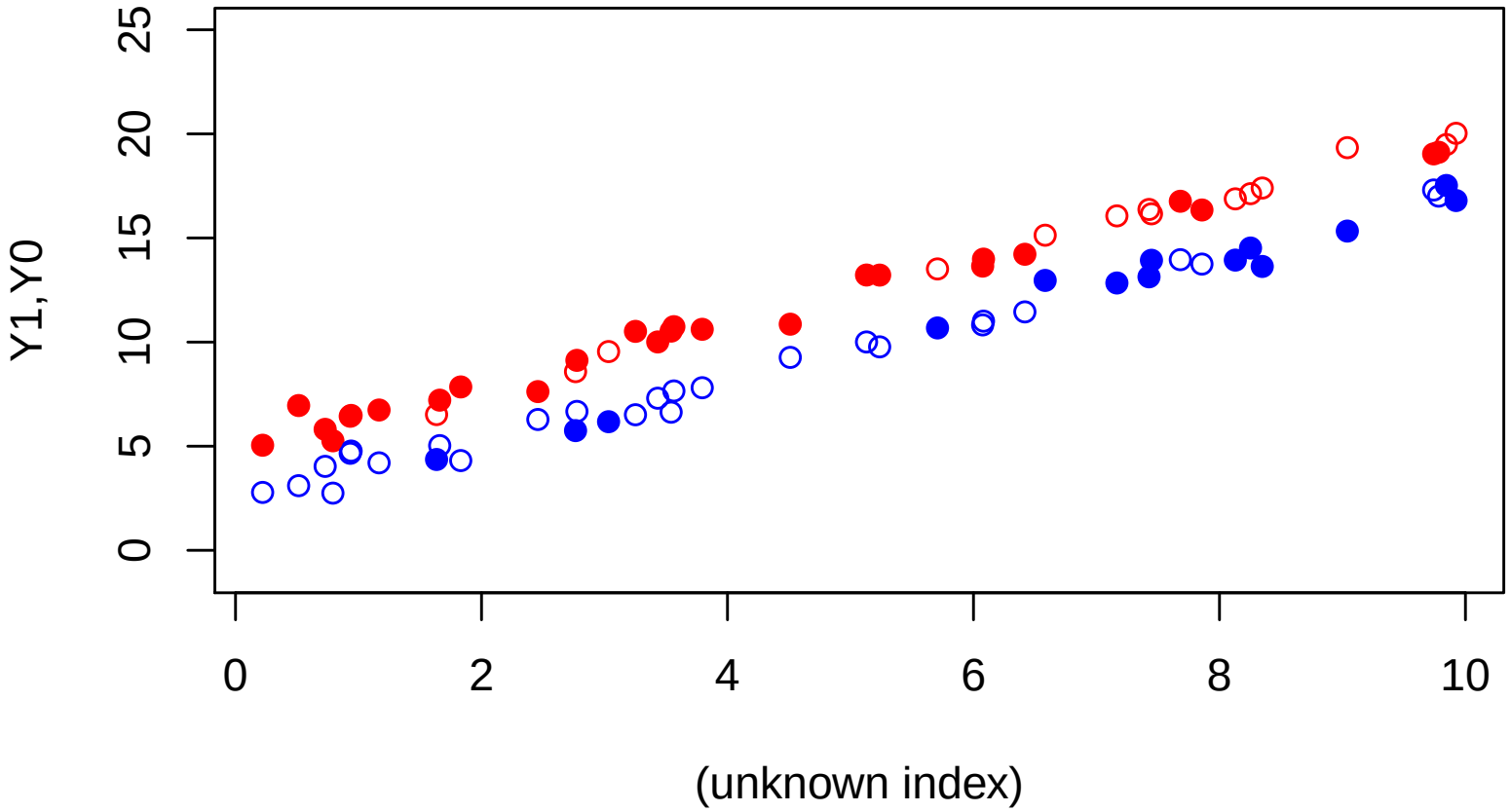

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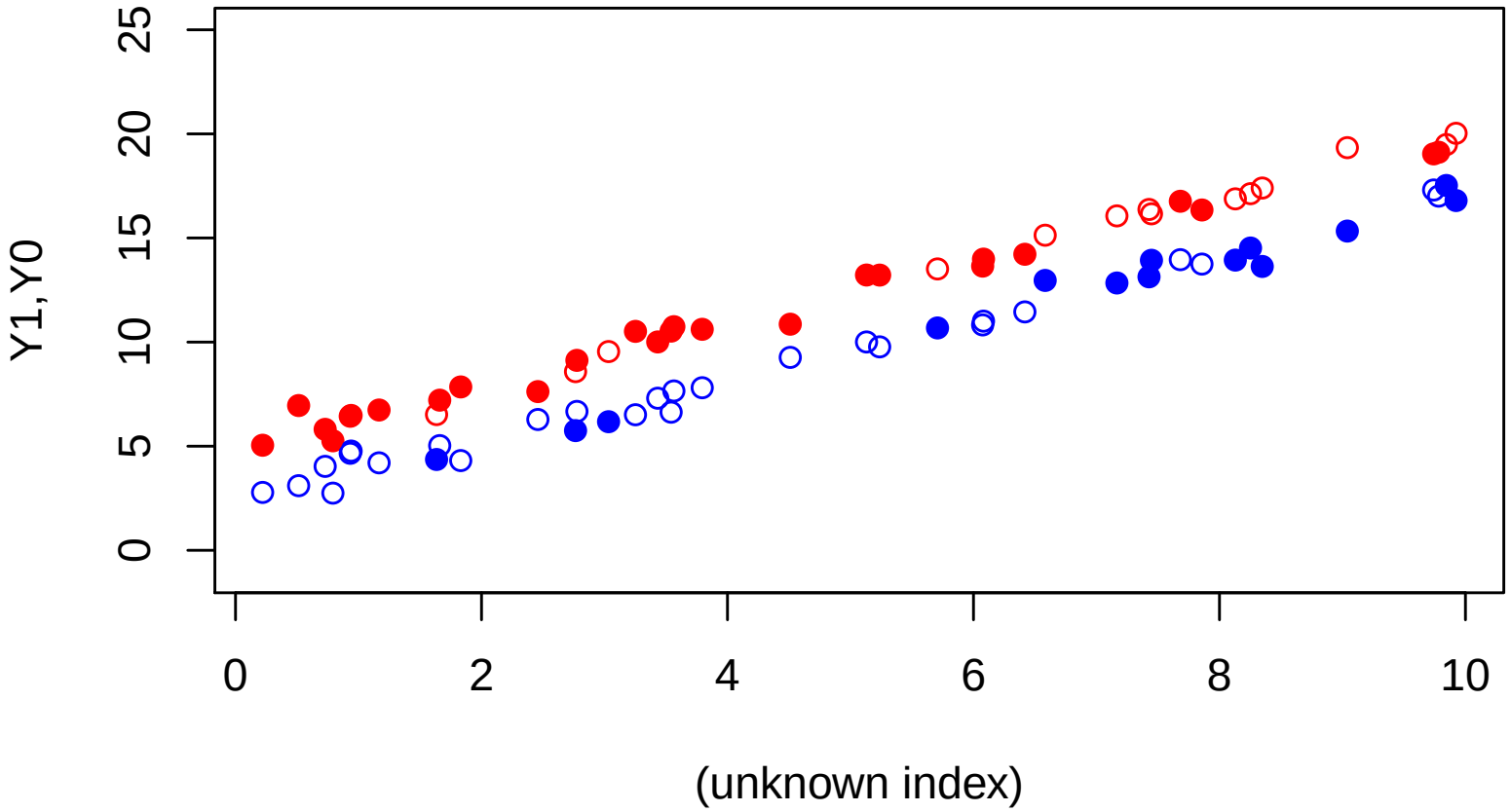

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$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{\text{DIM}} = -1.59 \neq \text{ATE} = 3$$

Yet More Brain Training: “science” tables

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
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(Note: average effect is positive, but not all τ_i are)

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Difference in Means

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What are the terms in blue & green?

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- **Next:** what has to be true about how D is assigned for the observed difference to recover a causal quantity? $\rightarrow$ randomization, selection-on-observables, and the rest of the course.